

Smart Manufacturing IoT-Based Predictive Maintenance & Anomaly Detection

Aomsin Nimsombun Colin Xiong Kaoklong Panitpongpun Daniel Benjamin

Group 22 — MANU 465, University of British Columbia

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Problem & Motivation

- **Context:** IoT sensors continuously monitor factory equipment (temperature, vibration, pressure, humidity)
- **Problem:** Most facilities still use reactive or scheduled maintenance
 - Unplanned downtime costs hundreds of thousands of dollars per hour
 - Unnecessary servicing wastes resources
- **Goal:** Predict maintenance needs from sensor data before failures occur
- **Secondary goal:** Detect anomalies without labeled data (unsupervised)

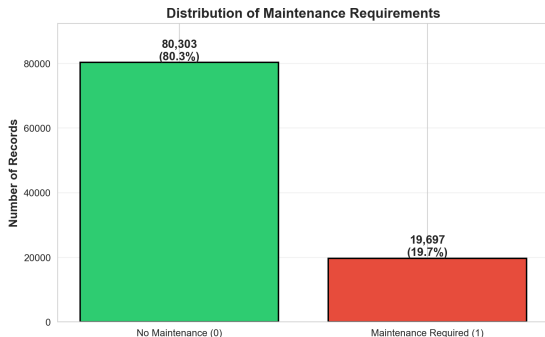
Dataset Overview

Smart Manufacturing IoT-Cloud Monitoring Dataset (Kaggle)

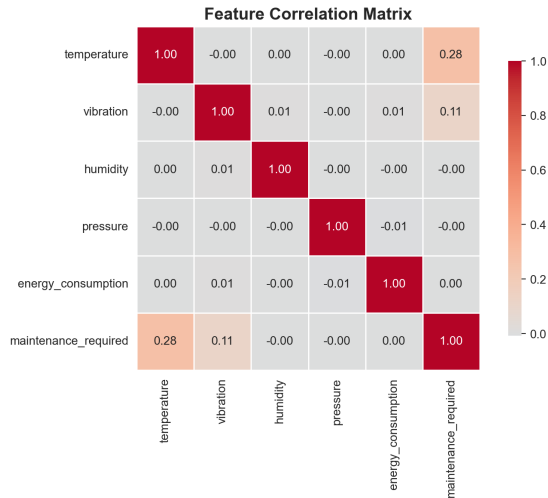
- 100,000 records, 50 machines
- 1-minute intervals over 69 days
- 13 columns total; 5 raw sensor features used
- Derived features excluded (data leakage)
- No missing values

Key correlations with target:

- Temperature: $r = 0.283$
- Vibration: $r = 0.113$



Correlation Heatmap



Weak individual correlations suggest **non-linear feature combinations** are important.

Preprocessing

- ❶ Dropped identifiers (`timestamp`, `machine_id`)
- ❷ Excluded derived features that encode the target (`machine_status`, `anomaly_flag`, `predicted_remaining_life`, `downtime_risk`, `failure_type`)
- ❸ Retained 5 raw sensor features
- ❹ Stratified 80/20 train-test split (80k / 20k samples)
- ❺ StandardScaler fitted **only on training data** (prevents leakage)
- ❻ `class_weight='balanced'` for imbalanced classes

Supervised:

- ① **Logistic Regression** — interpretable baseline with cross-validation
- ② **Random Forest** — non-linear ensemble with GridSearchCV tuning
- ③ **ANN (TensorFlow/Keras)** — deep non-linear modeling with Early Stopping

Unsupervised / Reduction:

- ④ **PCA** — dimensionality reduction, variance analysis
- ⑤ **Isolation Forest** — anomaly detection without labels

Evaluation metrics:

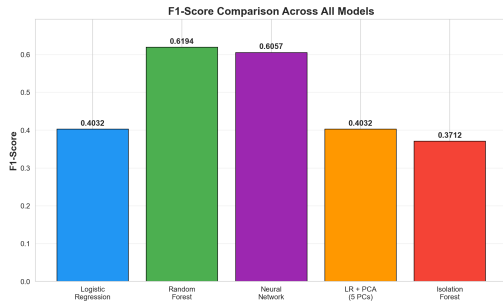
Accuracy, Precision, Recall, F1, ROC-AUC, Confusion Matrices

Input (5) $\rightarrow$ Dense(64, ReLU) $\rightarrow$ Dense(32, ReLU) $\rightarrow$ Dense(16, ReLU) $\rightarrow$ Dense(1, sigmoid)

- 3 hidden layers, Adam optimizer ($lr = 0.001$), binary crossentropy
- Early Stopping (patience = 10) with best-weight restoration
- Balanced class weights, batch size 256, 20% validation split

Model Comparison

Model	Acc.	Prec.	Rec.	F1	AUC
Naive Baseline	~0.80	—	—	—	—
Logistic Reg.	0.6352	0.2974	0.6255	0.4032	0.6929
Random Forest	0.8913	0.9989	0.4488	0.6194	0.7230
Neural Network	0.8831	0.9029	0.4557	0.6057	0.7250
LR + PCA	0.6352	0.2974	0.6255	0.4032	0.6929
Isolation Forest	0.7476	0.3644	0.3783	0.3712	0.6571



- Non-linear models (RF, ANN) achieve $F1 \approx 0.62$ vs. LR's 0.4032
- RF and ANN reach nearly identical AUC — performance ceiling with 5 raw sensors

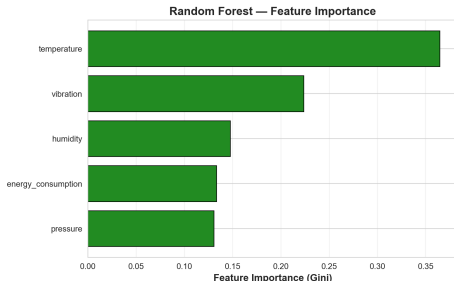
Key Results: Logistic Regression & Random Forest

Logistic Regression:

- Accuracy $0.6352 < \text{naive } 0.80$
- Recall 0.6255 ; precision 0.2974
- CV F1: 0.4091 ± 0.0060
- Linear boundary limits performance

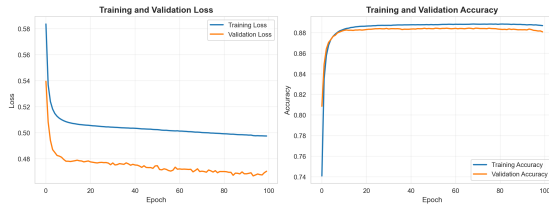
Random Forest (tuned):

- F1: $0.4032 \text{ (LR)} \rightarrow 0.6194$
- Precision $0.9989 \approx \text{perfect}$
- Recall 0.4488 ; misses $>50\%$ of failures
- temperature + vibration = 58% importance

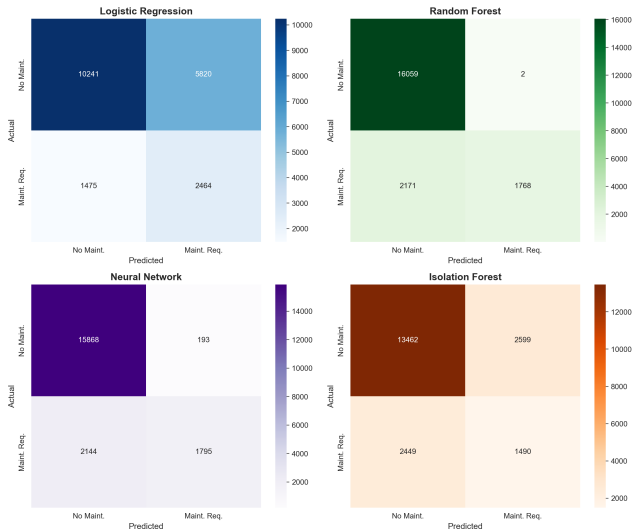


Key Results: Neural Network

- $F1 = 0.6057 \approx \text{RF's } 0.6194$
- Precision 0.9029, recall 0.4557
- 2144 missed failures, 193 false alarms
- Early Stopping at epoch ~ 90
- $AUC = 0.7250$, tied with RF

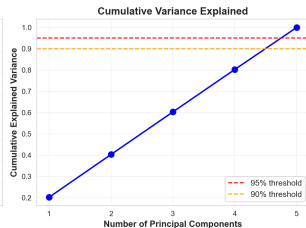
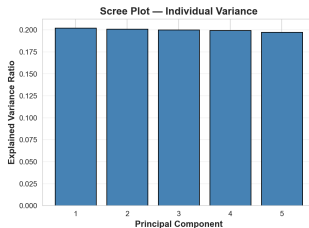


Confusion Matrices — All Models



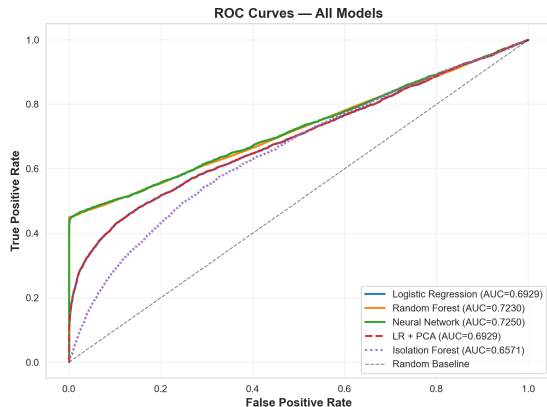
PCA — Variance Analysis

- All 5/5 components needed for 95% variance
- PC1: 20.2%, PC2: 20.1% — nearly uniform
- LR+PCA F1 = 0.4032 ($\Delta F1 = 0.0000$)
- Sensors are nearly orthogonal: no redundancy



Isolation Forest — Unsupervised Anomaly Detection

- $F1 = 0.3712$, $AUC = 0.6571$ — above random (0.5), real signal captured
- 2599 false alarms, 2449 missed failures — balanced error profile
- Label value gap: RF F1 0.6194 — IF F1 0.3712 = 0.25
- Viable first-line monitoring when labeled data is unavailable



Strengths & Limitations

Strengths:

- Five techniques across model families
- Rigorous evaluation (stratified split, CV, multiple metrics, naive baseline)
- Data leakage prevention
- Hyperparameter tuning & regularization
- Actionable feature importance

Limitations:

- Synthetic dataset — may not reflect real-world noise
- Only 5 raw sensor features; richer engineering possible
- No temporal modeling (records treated independently)
- Single dataset — generalizability unvalidated

Conclusion & Future Work

Key findings:

- RF achieved best F1 (0.6194) with near-perfect precision (0.9989); ANN matched its AUC (0.7250)
- Non-linear models improved F1 by $\sim 50\%$ over LR baseline (0.4032)
- Temperature (0.365) and vibration (0.224) account for 58% of RF's predictive power
- PCA: all 5 sensors are non-redundant (each $\approx 20\%$ variance)
- Isolation Forest captures real anomalous structure (AUC = 0.6571) without labels
- Performance ceiling at $F1 \approx 0.62$ suggests richer features are needed

Future work:

- LSTM / GRU for temporal degradation patterns
- Rolling-window feature engineering from raw sensors
- SHAP values for instance-level explainability
- Validate on real-world industrial datasets

Thank You

Questions?

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